

A Smart Computing Framework for AI-Driven Urban Drainage and Waterlogging Prediction

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Abstract—Urban waterlogging has emerged as a persistent challenge in rapidly growing cities due to increasing rainfall variability, unplanned urbanization, and aging drainage infrastructure. Traditional drainage monitoring and flood prediction approaches often rely on isolated data sources or computationally intensive simulations, limiting their responsiveness and scalability in real-world urban environments. This paper presents a smart computing framework for AI-driven urban drainage monitoring and waterlogging risk management that integrates IoT-based water-level sensing, GIS-based drainage mapping, and real-time meteorological data within a unified system architecture. The proposed framework emphasizes modular edge-cloud integration, multi-source data fusion, and decision-support capabilities to enhance situational awareness and support proactive drainage management. Rather than focusing on algorithmic benchmarking, the paper contributes a system-level design perspective that highlights architectural choices, data workflows, and operational considerations for smart city deployments. The framework is intended to serve as a scalable foundation for future experimental validation, optimization, and real-world implementation of intelligent and climate-resilient urban drainage systems.

Index Terms—Urban drainage systems, waterlogging risk management, Internet of Things (IoT), Geographic Information Systems (GIS), smart cities, artificial intelligence (AI), multi-source data fusion, edge-cloud computing, decision support systems.

I. INTRODUCTION

Due in large part to shifting rainfall patterns, fast urbanization, and antiquated drainage systems, urban waterlogging has become an increasingly widespread and enduring problem in cities all over the world [1]. With a large percentage of paved and impermeable surfaces, modern cities are particularly vulnerable; even light rainfall can now result in flooded streets, traffic jams, financial losses, and health hazards for the general public [2]. As climate change brings more heavy and unpredictable rainfall, events that were once thought to be extreme are happening more often [1]. Flooding is exacerbated in many urban areas through overburdened drainage systems, dwindling natural water bodies, and improper waste management.

Despite increasing use of artificial intelligence to forecast flooding, most of the existing systems are inefficient and ineffective due to their heavy reliance on either simulations or a single data source [3]. Conventional water management methods, despite their precision, even though hydrodynamic models are frequently necessitating substantial resources, sluggish, and reliant on data, therefore unsuitable for quick decision-making in busy metropolitan regions [4]. In the same way, AI models

that depend exclusively on Meteorological data and flood records frequently have information deficiencies, a deficiency in spatial information, and an incapacity to adapt to novel or severe climatic conditions [5]. The failure of multiple systems to effectively represent real-world conditions results in the inability to consider the complexity of how water moves through drainage systems or integrate real time information from IoT sensors [6].

The limitations of these systems result in reduced effectiveness in managing flooding and can reduce the overall long-term sustainability of urban areas [7]. Low-income communities with high levels of vulnerability are often disproportionately affected by waterlogged homes, contaminated drinking water and increased rates of disease due to flooding [1]. A multi-faceted approach that incorporates a comprehensive real-time system utilizing real-time water level monitoring using IoT technology along with GIS drainage mapping utilizing current weather data will improve forecast accuracy and allow for better decision making for emergency responders and urban planners to design more adaptive, sustainable, and resilient urban environments [8] [9].

The key contributions of this paper are as follows:

- The design of a unified AI-driven framework that integrates IoT sensing, GIS-based drainage mapping, and real-time meteorological data to support urban waterlogging risk monitoring and management.
- A modular edge-cloud system architecture that supports distributed data processing, scalable deployment, and real-time decision-support functionality for urban drainage applications.
- A qualitative design-level evaluation that illustrates how integrated smart drainage frameworks can address the limitations of traditional, isolated, and centrally managed drainage monitoring systems.

II. LITERATURE REVIEW

A great deal has been achieved in predicting and detecting urban flooding and obstruction in urban drainage systems due to recent improvements in Artificial Intelligence (AI) and Machine Learning (ML) technology [4]. Predictive models such as Random Forests, Long Short-Term Memory (LSTM)

Networks, and Ensemble Techniques are currently capable of generating fast and accurate predictions regarding water levels and the likelihood of a flood [7]. Compared with traditional hydraulic simulation-based approaches that often require large amounts of processing time and fixed data sets, these modern AI-based predictive models can quickly adapt to the ever-changing urban environment and its complexities [10]. Additionally, the combination of AI and physical simulation enables near-real-time forecasting of urban floods; allowing for proactive planning of maintenance activities and emergency responses [2]. However, while modern predictive models have many advantages, their predictive accuracy is highly dependent upon the quality, variability, and consistency of the input data [3].

The Internet of Things (IoT) has also created new possibilities for the continuous monitoring and management of Urban Drainage Systems, enabling city managers and engineers with real-time data to make informed decisions [6]. IoT devices that are networked together can monitor water levels, flow rates, debris accumulation, and automatically control pumps and valves to prevent an overflow in case of a storm event. Using Satellites, the City will have the ability to predict heavy rainfall events with unprecedented detail and lead times, and anticipate potential blockages, thus providing early flood warnings when combined with AI Analytics [5]. Applications in the real world have demonstrated that this combination can increase the cost-effectiveness and proactiveness of drainage management [9]. However, the majority of these implementations are still small-scale, with problems like erratic connectivity and challenges when trying to scale to larger, older, and more varied city infrastructures.

In order to comprehend how water flows through cities, Geographic Information Systems (GIS) are also essential [1]. GIS tools are useful for mapping drainage networks, identifying areas that are vulnerable to flooding, and analyzing the interactions between infrastructure, terrain, and land use [11]. GIS enhances spatial accuracy and assists in identifying potential waterlogging locations and times when combined with AI [12]. However, it is still rare that the full integration of GIS with AI analytics and real-time IoT data, which narrows down the possibility of truly comprehensive flood forecasting systems [10].

A multidisciplinary research approach that integrated different data types was used, such as meteorological data, hydrological patterns, spatial layouts, and real-time sensor readings—into unified systems in order to get around these constraints [7]. This model leverages multiple sources of data, compared to models that use one data source. The fusion of source data enables the making of more accurate, real-time, and context-aware predictions [4]. However, there are still issues in integrating these different data streams in a smooth manner, ensuring the quality of the data, and maintaining interoperability between systems operated by different organizations or built with various technologies.

Despite advancements, a large number of AI-based drainage solutions still depend on discrete datasets, which restrain

their ability to adapt and survive in operational situations [13]. There are very few frameworks that successfully combine IoT sensor feedback, GIS-based drainage mapping, and meteorological data onto one single, coherent platform [8]. Scaling and integration are difficult since there are fragmented data infrastructures and a lack of standardized protocols to exchange data [3]. Furthermore, long-term resilience and sustainability aspects such as infrastructure decay, the impacts of climate change, and the need for equitable access to flood protection—must be frequently reconsidered [7]. These needs must be addressed in order to develop urban drainage systems that are both adaptable and intelligent as they provide a foundation for sustainable and resilient urban growth within modern cities.

Existing studies have demonstrated the potential of artificial intelligence, IoT-based sensing, and GIS tools for urban drainage monitoring and flood risk assessment. However, most prior approaches focus on isolated components, such as data-driven prediction models, sensor-based monitoring, or spatial analysis, often implemented as standalone solutions. As a result, challenges related to system interoperability, real-time data fusion, deployment scalability, and operational decision support remain insufficiently addressed. Moreover, many proposed methods emphasize algorithmic accuracy without considering practical integration with urban infrastructure or adaptive system design. This paper addresses these gaps by proposing a unified, modular framework that integrates sensing, spatial analysis, and intelligent decision support within a coherent edge–cloud architecture tailored for smart city drainage applications.

III. PROPOSED METHODOLOGY

1. Source of Input Data

To develop a comprehensive understanding of urban drainage systems, the proposed framework is designed to utilize data from multiple heterogeneous sources, including meteorological, spatial, and sensor-based inputs:

- **Real-time meteorological data:** Data obtained from meteorological stations and forecasting services provide continuous information on rainfall characteristics such as duration and intensity. These inputs are used to support flood-risk assessment and situational analysis within the proposed framework [1].
- **GIS-based maps:** High-resolution Geographic Information System (GIS) maps are used to represent the drainage network, including components such as pipes, catch basins, and manholes, along with their spatial relationships to surrounding infrastructure elements such as roads and terrain features [11].
- **IoT water level sensors:** Smart sensing devices are incorporated to support continuous observation of water levels, flow rates, and abnormal fluctuations within drainage channels. These inputs are intended to assist in identifying potential conditions such as blockages or

saturated soils and to support responsive system operation under changing environmental conditions [6].

2. Architecture of AI Models

The system incorporates an AI-based analytical component designed to capture temporal rainfall patterns and their interaction with spatial characteristics of urban geography, supporting the assessment of water movement behavior within drainage networks.

- **Time-Based Patterns:** The temporal component is designed to process rainfall and water-level sequences using time-aware neural network structures, such as BiTCN and GRU, to support the analysis of short-term variations and longer-term temporal trends in hydrological conditions [8].
- **Space-Based Patterns:** GIS-derived spatial information is incorporated to represent the structure, capacity, and connectivity of the drainage network. This spatial context supports the analysis of how geographical configuration influences water movement and potential accumulation within urban areas [12].
- **Module Integration:** By integrating the temporal and spatial components, the framework supports location-aware flood-risk assessment by combining rainfall dynamics with drainage network characteristics across different urban zones [10].

3. Data Fusion Strategy

The framework integrates data from multiple heterogeneous sources, including sensor measurements, historical rainfall information, and drainage maps, to support a unified data representation for flood-risk assessment within urban drainage systems.

- **Feature-Level Fusion:** Feature-level data fusion is employed to align and combine heterogeneous inputs, enabling contextual analysis of how multiple environmental and infrastructural factors influence waterlogging risk across different locations and time periods [9].
- **Hybrid Ensemble Learning:** The framework conceptually combines data-driven AI components with established hydrologic modeling principles to support robust flood-risk assessment and to enhance consistency between analytical outputs and domain-specific drainage behavior [3].
- **Dynamic Model Updates:** The framework is designed to support iterative model updating in response to evolving weather patterns and infrastructure conditions by incorporating newly available data streams. This capability enables continued system adaptability and sustained decision-support functionality under changing urban environments [8].

4. Forecasting and Decision-Making Process

The proposed framework operates as a continuous, real-time decision-support pipeline designed to assist urban drainage monitoring and management:

- **Data Acquisition:** GIS-based drainage maps, IoT sensor readings, and live meteorological updates are contin-

uously collected, synchronized, and stored to support system-level analysis and situational awareness [6].

- **Preprocessing:** Incoming data streams are validated, cleaned, and spatially aligned using standardized reference systems to ensure consistency and reliability before being utilized within the analytical components of the framework [4].
- **Feature Engineering:** Relevant environmental and infrastructural attributes—such as precipitation levels, historical water observations, elevation, and proximity to drainage infrastructure—are derived to support contextual risk assessment and analytical interpretation within the system [1].
- **Model Inference:** The AI-based analytical component processes the integrated data streams to support spatially informed assessment of waterlogging risk across different drainage zones and urban regions [1].
- **Decision Support:** The framework provides visual dashboards, alert indicators, and contextual information to assist emergency responders and city planners in informed decision-making. The architecture is designed to support integration with automated drainage control mechanisms where operationally appropriate [2].
- **Continuous Learning:** The framework supports iterative updating of analytical components by incorporating newly available data, enabling continued adaptability to evolving environmental conditions and infrastructure changes [9].

Overall, the proposed framework provides a structured, data-driven approach for supporting resilient urban drainage management by integrating sensing, spatial analysis, and AI-based decision support within a unified system design.

5. Block Diagram

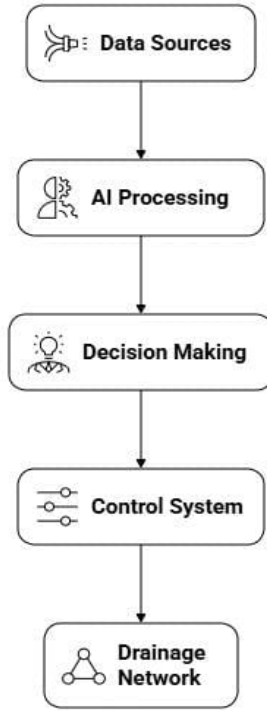


Figure 5.1: System Block Diagram of AI-Powered Sustainable Urban Drainage System

IV. SYSTEM ARCHITECTURE

The proposed AI-based sustainable urban drainage framework adopts a modular, multi-layered architecture that integrates real-time monitoring, analytical processing, and automated control components to support urban waterlogging management [9]. Each architectural layer is designed to perform specific functions while maintaining coordinated interaction with other layers to enable coherent and data-driven system operation.

1. Layer of Data Acquisition

The data acquisition layer forms the foundation of the system and is responsible for collecting relevant environmental and infrastructure-related data required for urban drainage analysis.

- **Meteorological Data Sources:** Meteorological stations and forecasting services provide continuous information on rainfall intensity, duration, and spatial distribution patterns [1].
- **IoT Sensor Network:** An IoT sensor network comprising ultrasonic water-level sensors, flowmeters, and water-quality sensors is deployed along drainage channels and catch basins to support real-time observation of flow conditions and water-level variations [6].
- **GIS Drainage Maps:** High-resolution GIS-based spatial maps provide geographical context by representing drainage network layouts, elevation profiles, and land-use characteristics relevant to urban water management [11].

2. Edge Computing Layer

The edge computing layer is designed to support localized data processing and preliminary analysis close to the data sources within the urban drainage infrastructure.

- **Edge Gateways and Microcontrollers:** The edge computing layer incorporates gateways and microcontroller-based devices, such as Raspberry Pi and ESP32 platforms, to perform preliminary data processing near the data sources. These operations include noise filtering, normalization, and basic feature extraction, which help reduce communication overhead and support efficient data transmission to higher-level analytical components [4].
- **Local AI Inference:** Lightweight AI-based analytical components, such as resource-efficient neural network models, are conceptually deployed at edge nodes to support preliminary pattern analysis and anomaly indication based on locally observed sensor data. This localized analysis is intended to enhance system responsiveness and contribute to higher-level decision-support functions [8].

3. Fusion Layer and Centralized Analytics

This layer is designed to integrate heterogeneous data streams and support centralized analytical processing for informed urban drainage management.

- **Cloud Platform:** Centralized cloud infrastructure is used to aggregate meteorological information, preprocessed edge data, and GIS-based spatial data, providing a scalable environment for advanced analytical and simulation-based processing [3].
- **Data Fusion:** The data fusion module integrates historical sensor observations with spatial and meteorological context to support comprehensive assessment of waterlogging risk and drainage behavior across urban regions [7].
- **Decision Support:** The decision-support component provides visualizations, status indicators, and contextual information to assist city officials in informed planning and maintenance-related decision-making. These outputs are intended to support proactive management of urban drainage infrastructure [2].

4. Layer of Control and Actuation

This layer is designed to utilize continuously updated system information to support responsive control actions and operational awareness within the urban drainage infrastructure.

- **Smart Pumps and Valves:** Automated pumping and valve mechanisms are conceptually integrated into the framework to support adaptive control of drainage operations based on system status information. These components are intended to assist in regulating flow conditions under varying environmental scenarios [9].
- **Dashboard and user interface:** A map-oriented user interface is designed to visualize water-level conditions, drainage-sensitive regions, and system status indicators, supporting situational awareness and informed operational oversight by system operators [6].

5. Layer of Feedback and Adaptation

This layer is designed to support long-term system adaptability, scalability, and operational robustness.

- **System Adaptation:** The framework supports iterative updating of analytical components by incorporating feedback from system observations and newly available data streams. This capability is intended to facilitate adaptability to evolving environmental conditions and infrastructure changes over time [8].
- **Scalability and Resilience:** The architectural design incorporates distributed computing principles and redundant communication mechanisms, such as LoRaWAN-based mesh networks, to support scalable deployment and resilient operation across large urban areas [4].

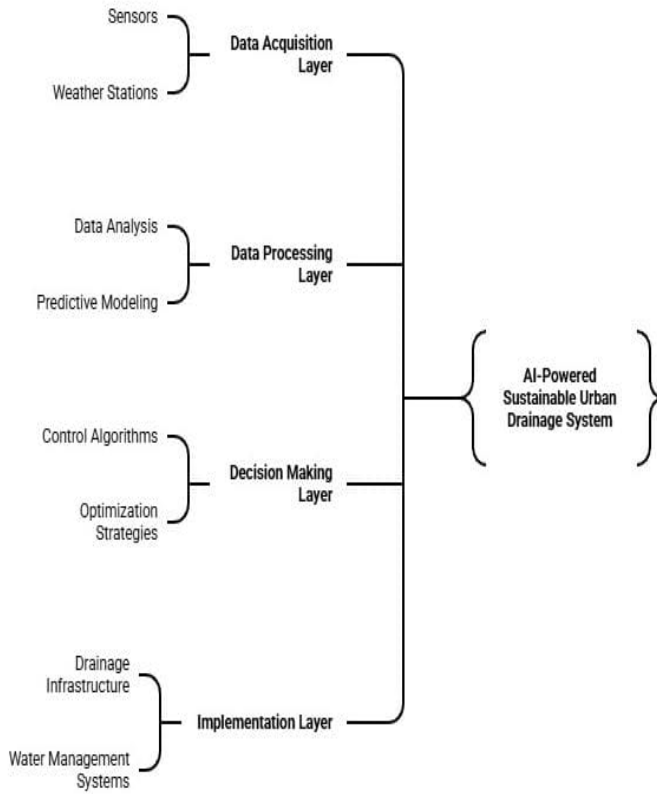


Figure 6.1: Conceptual system architecture illustrating the data flow and functional components of the proposed AI-enabled urban drainage framework

V. QUALITATIVE EVALUATION AND DESIGN DISCUSSION

The proposed multi-source AI-driven urban drainage framework is evaluated qualitatively through a design-oriented anal-

ysis, focusing on system architecture, data integration strategy, and operational feasibility rather than quantitative benchmarking. This evaluation approach is appropriate for early-stage system frameworks where the primary contribution lies in architectural integration and workflow design rather than in algorithmic optimization or dataset-specific performance.

A key strength of the proposed framework lies in its multi-layered edge-cloud architecture, which is designed to support timely monitoring and decision-making in complex urban drainage environments. By distributing data preprocessing and preliminary inference tasks to edge devices, the system architecture aims to reduce dependency on centralized processing and enable faster local responses to rapidly changing water-level conditions. This design choice addresses known latency and scalability challenges associated with traditional centralized SCADA-based drainage monitoring systems.

The integration of heterogeneous data sources, including IoT-based water-level sensing, GIS-derived spatial drainage information, and real-time meteorological inputs, enhances the contextual awareness of the system. Unlike isolated monitoring solutions that rely on single data streams, the proposed framework emphasizes feature-level data fusion to support more informed risk assessment and situational understanding. This integrated perspective is particularly valuable in urban settings where drainage behavior is influenced by both spatial infrastructure constraints and dynamic weather patterns.

From a system design perspective, the framework supports adaptive and extensible operation. The modular structure allows individual components—such as sensing infrastructure, analytical models, and decision-support interfaces—to be updated or replaced without requiring major architectural changes. This flexibility is essential for real-world deployment, where infrastructure heterogeneity, evolving sensor technologies, and city-specific requirements are common.

The inclusion of a decision-support layer further strengthens the practical relevance of the framework. By translating analytical outputs into visual maps, alerts, and actionable insights, the system is designed to assist city officials and emergency responders in proactive drainage management and flood-risk mitigation. Although automated actuation mechanisms are conceptually incorporated, their deployment and validation are intentionally positioned as future extensions to ensure system reliability and operational safety.

Despite these strengths, certain limitations are acknowledged at the framework level. The effectiveness of the proposed design is inherently dependent on data availability, sensor reliability, and communication stability, particularly in dense or infrastructure-constrained urban areas. Additionally, while the framework outlines the integration of AI-based predictive components, detailed experimental validation, performance benchmarking, and comparative analysis with baseline models remain outside the scope of the current study.

Overall, this qualitative evaluation demonstrates that the proposed framework offers a coherent, scalable, and adaptable system design for AI-enabled urban drainage monitoring. By addressing fragmentation across sensing, spatial analysis,

and decision support, the framework provides a structured foundation for future implementation, experimentation, and performance-driven optimization in real-world smart city environments.

VI. CONCLUSION

This paper presented a multi-source smart computing framework for AI-enabled urban drainage monitoring and water-logging risk management. By integrating IoT-based water-level sensing, GIS-derived spatial drainage information, and real-time meteorological data within a unified edge–cloud architecture, the framework addresses key limitations of fragmented and centrally controlled drainage monitoring systems. The proposed design emphasizes modularity, scalability, and contextual awareness to support proactive decision-making in complex urban environments.

Rather than focusing on algorithmic performance benchmarking, this work contributes a system-level perspective that outlines architectural components, data workflows, and operational considerations essential for implementing intelligent urban drainage solutions. The qualitative evaluation highlights how multi-source data fusion and distributed processing can enhance situational awareness and support adaptive drainage management in smart cities.

While experimental validation and quantitative performance analysis are beyond the scope of the present study, the proposed framework provides a structured foundation for future research, real-world deployment, and empirical evaluation. As urban flooding risks continue to intensify under climate variability and infrastructure stress, such integrated and adaptable system designs will play a crucial role in advancing sustainable and resilient urban water management.

VII. FUTURE WORK

The future scope of the framework consists of potential improvements, which are going to be the impact on the performance, the scalability, and the reliability of the framework. Remote sensing merged with rainfall as monitored by satellites. On the other hand, federated learning would permit the joint training of models in different cities without breaching their data privacy, whereas nowcasting could foresee the conditions of a larger area and give longer predictions. The powerful protocols like LoRaWAN mesh would be the great choice to allow the reliable data transmission and to strengthen the communication networks particularly during the emergency situations. Digital twins, or the virtual representations of the drainage systems, would be the great option to allow the simulations of different scenarios and the optimization of the systems before the actual implementation in the world. Moreover, the extension of the framework's cloud-edge hybrid architecture and modularity could not only be scalable to heterogeneous infrastructures and megacities, but also, the use of explainable AI techniques to enhance model interpretability would support regulatory acceptance and win stakeholder trust.

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